Project Proposal

Project: Sales Performance Analysis: Bike & Accessories Retail

Objective: To analyze the sales performance of a bike and accessories retailer selling in six countries, identify the products, categories, markets, and customer segments that drive revenue and profit, and develop a model that forecasts monthly sales. The project will also estimate expected revenue and order volume for the coming months, helping decision makers improve inventory planning, product and pricing focus, marketing strategy, and customer retention.

Description: Retail sales performance depends on product mix, pricing, customer behavior, seasonality, and market conditions across countries. The project uses a sales dataset made of three related tables: sales transactions (order number, product, customer, order, shipping, and due dates, quantity, price, and sales amount), products (category, subcategory, product line, and cost), and customers (country, gender, marital status, and birth date). It covers about 60,000 order lines from roughly 27,700 orders placed by about 18,500 customers between December 2022 and January 2026.

The project will model the three tables as a star schema, clean and prepare the data, and investigate sales trends across time, products, categories, countries, and customer segments. It will measure revenue, profit, average order value, order frequency, and shipping time, and identify the factors that contribute most to sales performance and customer value.

Based on historical sales, the project will build a forecasting model for monthly sales and estimate expected revenue and order volume for the coming months. All results will be presented in an interactive Power BI dashboard, together with a final report and business recommendations.

Tools & Design:

- Data understanding and data modeling.
- Data cleaning and preprocessing.
- Exploratory and diagnostic analytics.
- Forecasting and predictive analytics.
- Interactive Power BI dashboard development and business communication.

Technologies to Use:

- SQL: Data modeling, extraction, transformation, cleaning, aggregation, and analytical queries.

- Python: Data preprocessing, exploratory analysis, statistical analysis, forecasting, and model evaluation.
- Python libraries: pandas, NumPy, Matplotlib, and scikit-learn.
- Power BI: Interactive dashboard, KPI reporting, trend analysis, forecasting visualization, and business storytelling.
- DAX: Measures, calculated metrics, KPI logic, and analytical calculations within Power BI.
- Microsoft Excel: Optional supporting tool for quick validation and data inspection.

Team Members & Responsibilities:

1- Anas Gaber — Star-schema data model, SQL structure, SQL queries, data validation, and DAX support.

2- Yamen Amr — Data cleaning, preprocessing, data quality checks, and documentation.

3- Asmaa Mokhtar — Analysis questions, KPI calculations, exploratory analysis, and dashboard planning.

4- Hassan Alaa — Product, market, and customer-segment analysis, business insights, customer-value analysis, and recommendations.

5- Yasmin Mahmoud — Forecasting, model evaluation, monthly sales prediction, Power BI dashboard development, and presentation coordination.

Week 1: *Data Model, Data Cleaning, and Preprocessing*

- Understand the dataset structure, business context, and meaning of each field across the sales, products, and customers tables. Responsible: All team members.
- Identify the keys and relationships linking sales to products and customers, and define the main dimensions and measures. Responsible: Anas Gaber.
- Design a star-schema data model with a sales fact table, product and customer dimension tables, and a date table. Responsible: Anas Gaber.

- Inspect data types, missing and “n/a” values (such as missing order dates, unknown countries, and uncategorized products), duplicate records, invalid date sequences, and outliers in price, quantity, and sales amount. Responsible: Yamen Amr.
- Clean and preprocess the data using SQL and Python, and derive new fields such as customer age, age group, profit, and shipping time. Responsible: Anas Gaber and Yamen Amr.
- Create a documented preprocessing notebook explaining cleaning decisions and transformations. Responsible: Yamen Amr.
- Validate the cleaned data and confirm that it is suitable for analysis and forecasting. Responsible: Anas Gaber and Yasmin Mahmoud.

Deliverables:

- Documented star-schema data model.
- Cleaned and analysis-ready tables.
- SQL scripts for data extraction, transformation, and cleaning.
- Data preprocessing notebook.
- Data dictionary describing the fields of all three tables.
- Data quality summary and validation results.

Week 2: *Analysis Questions and Exploratory Analytics*

- Determine the main business questions that can be answered using the available dataset. Responsible: Asmaa Mokhtar, with input from all team members.
- Identify key KPIs, such as total revenue, total profit, profit margin, number of orders, average order value, order frequency, and average shipping time.
- Analyze sales trends by month, quarter, and year, and by category, subcategory, product, and country.
- Identify the products, categories, customers, and countries with the highest and lowest sales and profit.
- Segment customers by age group, gender, marital status, and country, and compare their purchasing behavior and value.
- Use SQL and Python to calculate metrics, aggregate data, and generate visualizations. Responsible: All team members.
- Prioritize meaningful findings and convert them into business-oriented insights. Responsible: Asmaa Mokhtar and Hassan Alaa.

Deliverables:

- Documented list of analysis questions.
- SQL queries answering the selected questions.
- Exploratory data analysis notebook.
- KPI calculations and supporting visualizations.
- Analysis of major sales patterns, product performance, and customer segments.
- Initial business insights and recommendations.
- Shortlist of variables to be used in forecasting.

Week 3: *Forecasting and Predictive Analytics*

- Determine forecasting questions based on the dataset and findings from Week 2.
- Develop a forecasting approach for predicting monthly sales. Responsible: Yasmin Mahmoud.
- Aggregate sales into a monthly time series and prepare time-based features such as month, quarter, trend, and lagged sales.
- Compare suitable forecasting or machine-learning approaches, depending on the length and structure of the sales history.
- Evaluate model performance using appropriate metrics and time-based validation.
- Estimate expected revenue and order volume for the coming months based on the predicted sales.
- Break down the forecast by product category where the data supports it, and state all assumptions clearly.
- Create visualizations showing actual versus predicted sales and expected revenue.

Deliverables:

- Documented forecasting questions.
- Forecasting methodology and feature-selection explanation.
- Trained forecasting model.
- Model evaluation results.
- Monthly sales forecast.
- Estimated revenue and expected order volume.
- Actual-versus-predicted visualizations.
- Forecasting notebook and documented assumptions.
- Limitations and recommendations for improving the model.

Week 4: *Power BI Dashboard, Final Report, and Presentation*

- Design the overall dashboard structure and determine the most important pages and visualizations.
- Build an interactive Power BI dashboard containing KPIs, sales trends, category and product performance, country breakdowns, customer segments, and forecasting results.
- Create a Power BI data model and DAX measures required for KPI calculations and dashboard analysis.
- Add filters and interactive controls for dimensions such as date, country, category, product, and customer segment.
- Integrate the final analytical and forecasting outputs into the Power BI report.
- Create a management-focused view showing the main sales drivers, the monthly sales forecast, and estimated revenue.
- Validate dashboard calculations, visual consistency, and usability.
- Prepare the final report and presentation, including the business problem, methodology, data preparation, analysis, forecasting model, results, limitations, and recommendations.
- Rehearse the final presentation and ensure that each team member can explain their contribution.

Deliverables:

- Interactive Power BI dashboard.
- Power BI report and data model with DAX measures.
- Power BI dashboard documentation and KPI definitions.
- Final analytical and forecasting results.
- Final report.
- Final presentation.
- Business recommendations for increasing revenue, improving margins, and growing customer value.
- Complete project files, including SQL scripts, Python notebooks, cleaned datasets, and Power BI files.